Bridge Day 7 INSTRUCTOR KEY

Ordered Validation, Priority, and Nested Decisions

Learning sequence: Predict - Trace - Explain - Test - Interpret

Tuesday, July 14, 2026 • 20 points • target 16 minutes

Name		**UIN**		**Section**	

Assessment boundary and evidence source

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Cumulative foundations: input conversion, Boolean evidence, invalid-first branching, nested decisions, f-strings

Scaffold level: complete priority-sensitive classifier

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.14, 18.15, 18.16, 18.17.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace a priority-sensitive validation chain. - 5 points

```
quantity = -2

if quantity < 0:
    message = "invalid"
elif quantity <= 10:
    message = "small"
elif quantity <= 25:
    message = "medium"
else:
    message = "large"

print(message)
```

Question: Name the first matching branch and explain why a negative value cannot reach small.

Model response: quantity $<$ 0 is True, so invalid is assigned and the rest of the chain is skipped. A single if/elif chain preserves priority; exact output is invalid.

Item 2. Trace separate cost and classification decisions. - 5 points

```
mass = 28
purity = 90
rush = "yes"
cost = 12 + 1.8 * mass

if rush == "yes":
    cost = cost + 6

if purity < 85 or mass > 25:
    label = "hold"
else:
    label = "accepted"

print(label, cost)
```

Question: Show cost state before and after rush, then evaluate both classification components and give exact output.

Model response: Base cost is 12 + 1.8 * 28 = 62.4; rush raises it to 68.4. purity $<$ 85 is False, but mass $>$ 25 is True, so the or condition assigns hold. Exact output: hold 68.4.

Complete program - 10 points

- Read score as an integer and missing as text.
- Print invalid when score is outside 0 through 100.
- For valid input, print incomplete when missing is exactly yes, regardless of score.
- Otherwise print distinction for score at least 85, pass for score at least 70, and review for every lower valid score.
- Print exactly Result: followed by the label.

Program workspace

Model program

```
score = int(input())
missing = input()

if score < 0 or score > 100:
    print("invalid")
elif missing == "yes":
    print("Result:  incomplete")
elif score >= 85:
    print("Result:  distinction")
elif score >= 70:
    print("Result:  pass")
else:
    print("Result:  review")
```

Test input, expected result, and why this test matters

Reasoning and test evidence Invalid data has first priority. Among valid records, missing work has priority over the numeric labels, so a high score cannot bypass incomplete. Protective tests include 101/no -> invalid, 88/yes -> incomplete, 88/no -> distinction, 70/no -> pass, and 69/no -> review.
Scoring guidance: 2 input/domain, 2 missing priority, 3 descending score branches, 2 exact output and mutual exclusion, 1 boundary tests.

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.